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Research Topic: PROK2 and FGFR1 Mutations Responsible for HPG Disruption leading to Kallmann Syndrome in Women By Ashni Bhupathiraju

PROK2 and FGFR1 Mutations Responsible for HPG Disruption leading to Kallmann Syndrome in Women By Ashni Bhupathiraju

Abstract

Kallmann Syndrome (KS) has many underlying mechanisms, such as genetic mutations that cause neuroendocrine dysfunction. These mutations are associated with symptoms such as anosmia and hypogonadism, hence leading to infertility and loss of smell in men and women. However, the least understood aspect is how the Gonadotropin-releasing hormone (GnRH) neuron and hypothalamic-pituitary-gonadal (HPG) axis pathways could ultimately converge to cause GnRH dysfunction and female infertility. Although FGR1 and PROK2 are well-established as separate causative agents in KS, they likely work together across multiple stages of GnRH neuronal development. Understanding the mechanisms of these mutations and their effect on neuroendocrine dysfunction is crucial in decoding the link between GnRH development and hormone secretion. This disruption not only causes infertility and anosmia in women, but such insights would also deepen our understanding of reproductive neuroendocrinology and can inform the design of more targeted treatments for infertility. To address this gap, this review synthesizes current evidence on PROK2 and FGFR1 mutations, guided by the central question: How do FGFR1 and PROK2 mutations functionally converge to disrupt GnRH neuronal development and neuroendocrine signaling, contributing to female infertility in Kallmann syndrome?

Research Problem

Infertility, defined as the failure to achieve pregnancy after 12 or more months of regular unprotected sexual intercourse, affects nearly 1 in 7 couples worldwide, who experience the inability to conceive naturally (Sang et al., 2023). Kallmann Syndrome (KS) is a rare heterogenous genetic disorder in which affected individuals experience delayed or absence of puberty and reduced or absence of smell, leading to symptoms including anosmia and hypogonadism (Kumar Yadav et al., 2025). In women, the causation of certain infertility includes endopathies such as hypothalamic amenorrhea and hyperprolactinemia (Concepción-Zavaleta et al., 2023) however, neuroendocrine causes still remain underrepresented. FGFR1 and PROK2 stimulate GnRH release and trigger the activation of the mammalian reproductive axis. The hindrance of normal episodic GnRH secretion is thought to be the primary cause of KS, as these neurons play a major role in the regulation of the HPG axis (Young et al., 2019). The HPG axis which is responsible for puberty, sexual maturation, and reproductive abilities (Collado-Sole et al., 2026), connects the brain, pituitary gland, and reproductive organs. Hence, a disruption to the HPG axis can lead to several reproductive constraints. These disruptions may transpire through a series of gene mutations, namely FGFR1 and PROK2. With one or more of these mutated genes present within an individual, GnRH dysfunction is likely to take place. As GnRH neurons act as the fundamental regulators of the reproductive axis, their dysfunction is detrimental to the overall reproductive health of women.

GnRH neurons that migrate to the hypothalamus during embryonic development are activated by Kisspeptin neurons which secrete GnRH hormones that control the release of reproductive hormones (FSH and LH) through pulses and surges (Herbison, 2016). Although existing literature highlights the form and function of GnRH neurons after birth, the importance of their role during embryonic development receives comparatively less attention. As the FGFR1 gene plays an essential role in development of GnRH neurons (Yellapragada

et al., 2022), mutations in this gene cause both forms of isolate GnRH deficiency (IGD), accounting for approximately 10% of the GnRH deficiency cases (Miraoui et al., 2011). Mutations in FGFR1 and other genes such as KAL1 account for an estimated 10% of KS cases; therefore, a second gene that is investigated is PROK2 (Bianco & Kaiser, 2009). The absence of PROKR2, a primary G-coded receptor for PROK2, in mice has presented abnormal olfactory function and GnRH neuron migration (Martinez-Mayer & Perez-Millan, 2023). This indicates PROK2 as a primary gene associated with KS.

Over the past couple of decades, it has been established that mutations in PROK2 and FGFR1 are primary factors leading to KS. While studies examine the role of these two genes as causative genes in KS separately, there is minimal understanding of their synthesis and how they converge to bring upon symptoms of KS through the disruption of the HPG axis, especially in females. This review bridges the connection between both mutations and their collective role in the disruption of the HPG axis, causing KS. While reduced FGFR1 signaling triggers failed neurogenesis (Chung & Tsai, 2023), PROK2 affects the signaling pathways and migration of GnRH neurons. When understanding the converging signalling pathways of both FGFR1 and PROK2 genes, it provides insight into how the pathways provide complementary explanations of how GnRH functioning can be disrupted through different stages, leading to KS. Examining the convergence of both genes is instrumental in providing a biological insight into neuroendocrine infertility and advancing reproductive medicine in the future.

Methodology

To explore and deeply examine the relationship between the pathways that potentially converge between PROK2 and FGFR1 genes, contributing to Kallmann Syndrome, a qualitative literature-based approach was effectively utilized. As this review specifically focuses on two primary genes, along with their roles in GnRH development and migration, existing literature was thoroughly studied, analyzed, and later synthesized to produce the suggestions portrayed through this paper.

To acquire relevant and reliable research, academic databases including PubMed, Google Scholar and Frontier were used. Furthermore, the searches incorporated terms relating directly or indirectly to the main topic. Key words that were used included FGFR1, PROK2, Kallmann Syndrome, HPG axis, neuroendocrine dysfunction, and female infertility. With these search engines, a plethora of papers ranging from peer-review to experimental studies to review articles were found and utilized in the research. Additionally, relevant publications also produced reference lists which were extensively examined to identify additional studies that may be beneficial.

After obtaining the existing literature and information, the carefully selected literature was analyzed to determine whether evidence pointed towards a functional relationship between both signalling pathways being investigated. Seemingly, evidence acquired from studies of FGFR1 and PROK2 were compared on the basis of their effects on GnRH neuron development, migration, and signalling. Lastly, evidence was synthesized to observe how disruptions in said pathways may possibly converge on impaired GnRH signaling and HPG axis function, ultimately contributing to female infertility through Kallmann Syndrome.

To end, the analysis highlighted evidence from a vast array of experimental choices including animal-based, human-based, and laboratory experiments. Since this study

represents a collection and synthesis of existing literature rather than original experiments, conclusions were made while keeping in mind external limitations and methodology.

The Biological Basis of Fertility

Reproduction is vital for the survival and continuation of species. This life process is regulated and maintained by the HPG axis which consists of the anterior hypothalamus, pituitary gland, and gonads. To ensure the undisturbed function of this axis, GnRH neurons connect the neuronal pathways of the brain and send projections (Xie et al., 2022) through pulsatile secretion which stimulates the release of FSH and LH (Stamatiades & Kaiser, 2018). In standard female fertility, puberty takes place and, as a result, ovulation occurs. Ovulation, defined as the rupture of the dominant follicle of the ovary, releases an egg into the abdominal cavity. This process is governed by two primary reproductive hormones, FSH and LH. Roughly 1000 primordial follicles begin this process of maturation with the granulosa cell layer increasing in size, and production of estrogen commencing through FSH stimulation. FSH plays a role in beginning estrogen synthesis; however, this is later handled by granulosa cells as an autonomous process. As the cycle of follicle development progresses, one follicle called the "dominant follicle" secretes estrogen which wields negative feedback on FSH, hence causing all other follicles to "turn off". Continuing, the role of LH is mainly exhibited on day 14 of the menstrual cycle. As estrogen levels are high, it leads to an increased frequency of GnRH secretion, triggering a surge in LH levels. With LH levels high, intrafollicular proteolytic enzymes increase, weakening the walls of the ovary and allowing for the follicle to be released (Holesh et al., 2026). For couples attempting to conceive, this time period would be optimal as the follicle is exposed. Due to these governing hormones being dependent upon GnRH development and migration, when there is a disruption to the HPG axis due to genetic mutations, it often leads to KS in females.

As discussed, successful fertility cannot be attributed to a single hormone alone. Rather, there is a range of hormones that vary between males and females to facilitate conception. Migration of GnRH neurons through the neuronal pathways is divided into four distinct stages. These four stages demonstrate what "normal" GnRH functioning consists of. To begin, after birth in the olfactory placode, the next stage details a specific cue that guides the neurons across the cribriform plate. After this crossing, they migrate towards the hypothalamus, finally detaching from their axonal guides and dispersing throughout the hypothalamus (Wierman et al., 2011). Once established in the hypothalamus, GnRH neurons become the primary regulators of reproductive neuroendocrine signaling in the HPG axis.

Evidence supports that it's crucial for GnRH to successfully reach the hypothalamus, so the function of the HPG can be systematic. However, these stages of GnRH migration are only possible with the signals of guidance molecules including PROK2; therefore, a disruption to this molecule has profound consequences for the reproductive axis as a whole. Moreover, FGFR1 protein is a transmembrane receptor that plays an essential role in the development of GnRH as well as the migration of these neurons from the nasal placode.

FGFR1 Mutations and their Role in Neuroendocrine Dysfunction

Fibroblast growth receptor 1 (FGFR1) is a receptor tyrosine kinase (RTK). Variants of this protein have been identified in a majority of patients with KS. To detail, FGFR1 regulates many functions and processes including cell proliferation, differentiation, and migration of GnRH neurons. To carry out these processes, FGFR1 protein is encoded by the FGFR1 gene (Soejima et al., 2025).

FGFR1 not only regulates fetal development, it also plays a major role in the development of the nasal placode. The nasal placode, a region that brings about olfactory sensory neurons, infolds to form the primary olfactory epithelium and vomeronasal organ (structures where GnRH neurons begin to migrate from). The olfactory epithelium is a specialized tissue in the nasal cavity that is responsible for detecting odors, and the vomeronasal organ is another small sensory organ located in the nasal septum. After the odor molecules pass through these organs, they reach the olfactory bulb (OB) in the brain that processes the smell. FGFR1 mutations have been identified to disrupt this function. This is validated by studies conducted on mice which show that loss of FGFR1 results in disruption of olfactory bulb development and reduction in GnRH neuron count (Cho et al., 2019).

Moreover, the connection between FGFR1 disruption and irregular reproductive function is specifically important to female infertility through KS because of FGFR1's major role in GnRH development. Overall, a disruption to FGFR1 neuronal signalling can meddle with the growth and movement of GnRH neurons, reducing the GnRH count that is established within the hypothalamus. Because of GnRH neurons' role in regulating the production and secretion of FSH and LH, this reduced neuronal population is likely to weaken downstream HPG function. To regulate specific reproductive hormones, GnRH binds to a membrane receptor on the pituitary gonadotrophs which stimulates the secretion of the intended hormone (FSH) (Kaprara & Huhtaniemi, 2018). This provides a link between GnRH and hormone secretion, governed by FGFR1. Thus, the phenotype of KS can be understood through the downstream process of HPG disruption.

GnRH migration via PROK2 Signaling

While FGFR1 plays a fundamental role in the successful development of both the OB and GnRH neurons, orderly reproductive function also requires undisturbed migration of GnRH neurons from the nasal placode to the hypothalamus during the embryonic stage. To understand the importance of this migratory pathway, the ligand-receptor relationship between PROK2 and PROKR2 should be understood. Prokineticin receptor 2 (PROKR2) encodes a 384 amino acids long G-protein coupled receptor (GPCR) that can bind with PROKR1 and PROKR2. As both share 85% of the same genetic sequence, they share ligands, referred to as PROK1 and PROK2. When these ligands are bound to PROKR2, the GPCR leads to the activation of several signaling pathways. As PROKR2 shows high expression in the OB, it is considered essential for the development of the olfactory system and migration of GnRH (Martinez-Mayer & Perez-Millan, 2023).

To begin, GnRH neuronal migration follows a complex pathway commencing from their birth in the nasal placode to the mediobasal hypothalamus, where they complete their differentiation (Bassi et al., 2020). This migration, however, is only possible with the signalling of PROK2. To study the mechanism behind PROK2 causing neuroendocrine dysfunction and disruption to the HPG axis, studies were conducted on mice. Studies conducted in mice null for PROKR2 presented a loss of a majority of GnRH neurons in the forebrain (in adults) and they had also formed complicated webs of olfactory vomeronasal axons. This confirms the taxing neuroendocrine effects that PROK2 and GnRH dysfunction have on mice and humans.

To conclude, mutations that alter the ligand-receptor relationship between PROK2 and PROKR2 can disrupt signalling far beyond GnRH migration. With disturbance to the neuronal pathways connecting olfactory regions to the hypothalamus during embryonic development, the entire neuroendocrine signalling in female fertility is compromised.

Together, with KS-like phenotypes presented in PROKR2 knockout mice, it can be strongly argued that PROK2 is a primary protein involved in Kallmann Syndrome (Dodé et al., 2006).

Link between Genetic Mutations and Female Infertility: Kallmann Syndrome

Although mutations in FGFR1 and PROK2 are two significant factors leading to KS; there are more than 30 other genes that are associated with the syndrome, in which phenotypic correlation and novel mutations have been identified (Chu et al., 2023). For this reason, it is essential to understand that a mutation in one gene is not guaranteed to cause the same phenotypic symptoms in all affected individuals. The importance of understanding the convergence of more than one gene mutation is crucial, as it has been recently proven that KS is not just a monogenic Mendelian disease, rather it is oligogenic in nature (more than one gene is involved).

After identifying the connection between both genes, the understanding of their neuronal pathways "crossing" is essential. Crossing, in this case, refers to the functional convergence between the neuronal signalling pathways of both genes in which this overlapping causes the influences in the developmental process of the GnRH neurons. Considering phenotypes from studies using PROKR2 or PROK2 deficient mice and the likely effects in human mutations on protein synthesis, these mutations are predicted to be loss-of-function mutations as well. As KS results from insufficient signalling through either FGFR1 or PROKR2, it is expected that both of their signaling pathways interfere at some level. This interference could possibly be the activation of the mitogen-activated protein kinase pathway (Dodé et al., 2006). This argues that the signalling pathways of FGFR1 and PROK2 converge functionally on the development of the GnRH neuronal system to cause KS in females.

Together, both genes play different roles but functionally converge to alter the development and migration of olfactory and GnRH neurons, ultimately causing severe disruptions to the HPG axis. Through low GnRH neuron count and an underdeveloped olfactory bulb, the HPG can not function systematically, leading to Kallmann Syndrome (anosmia and CHH).

Conclusion

Overall, the evidence synthesized and reviewed suggests that Kallmann Syndrome is caused by a multitude of gene mutations, more specifically PROK2 and FGFR1. As these genes play their individual roles in disrupting the HPG axis, their convergence becomes evident through their neuronal pathways and phenotypic symptoms which they cause, including CHH and anosmia. This neuroendocrine dysfunction that leads to insubstantial secretion of female reproductive hormones is the primary cause of Kallmann Syndrome, ultimately leading to infertility in women. As FGFR1 and PROK2 both contribute to different aspects of GnRH and olfactory neuron dysfunction, the disruption of either of their pathways is likely to interfere with neuronal signalling.

As studies of Kallmann Syndrome, particularly associated with these two genes, advance, the likelihood of understanding the deeper mechanisms behind these mutations will increase. With evidence from studies consisting of affected female patients increasing, it is possible for a complex insight of these genes to be delivered. To conclude, further research into how combinations of genetic variants affect GnRH neuron development and neuroendocrine signalling could provide break-through evidence to advance the field of neuroendocrine infertility.

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